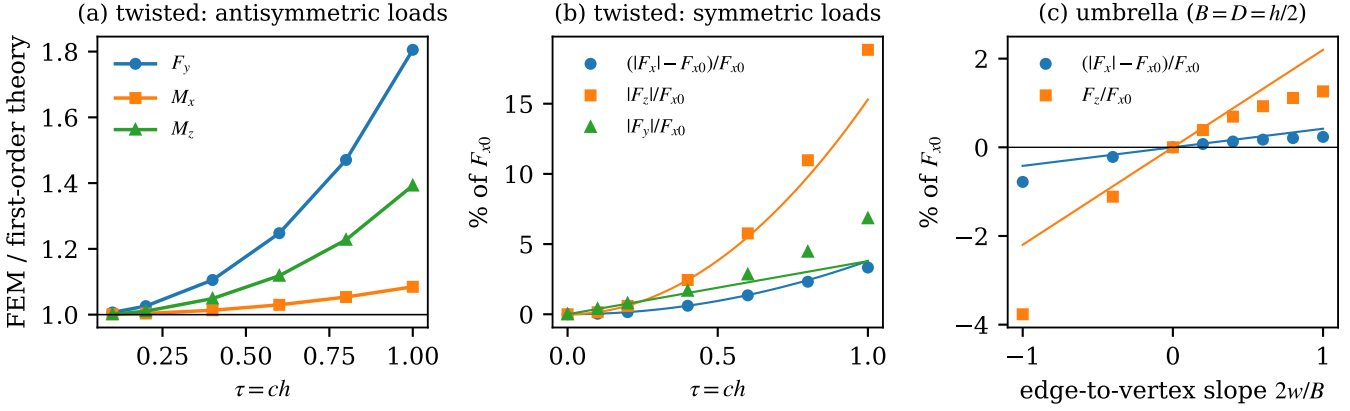
(a) twisted: antisymmetric loads
FEM / first-order theory
F_y
M_x
M_z
1.8
1.6
1.4
1.2
1.0
0.25
0.50
0.75
1.00
$\tau = ch$
(b) twisted: symmetric loads
% of F_{x0}
$(|F_x| - F_{x0})/F_{x0}$
$|F_z|/F_{x0}$
$|F_y|/F_{x0}$
15
10
5
0
0.0
0.5
1.0
$\tau = ch$
(c) umbrella ($B = D = h/2$)
% of F_{x0}
$(|F_x| - F_{x0})/F_{x0}$
F_z/F_{x0}
2
0
−2
−4
−1
0
1
edge-to-vertex slope $2w/B$